5 DOF Articulated Robotic Arm

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Subject Matter Expert – Dr. Qureshi

The University of Texas Rio Grande Valley

Manufacturing & Industrial Engineering

In Collaboration With: Los Alamos National Laboratory, TechSource, Savannah River Nuclear Solutions

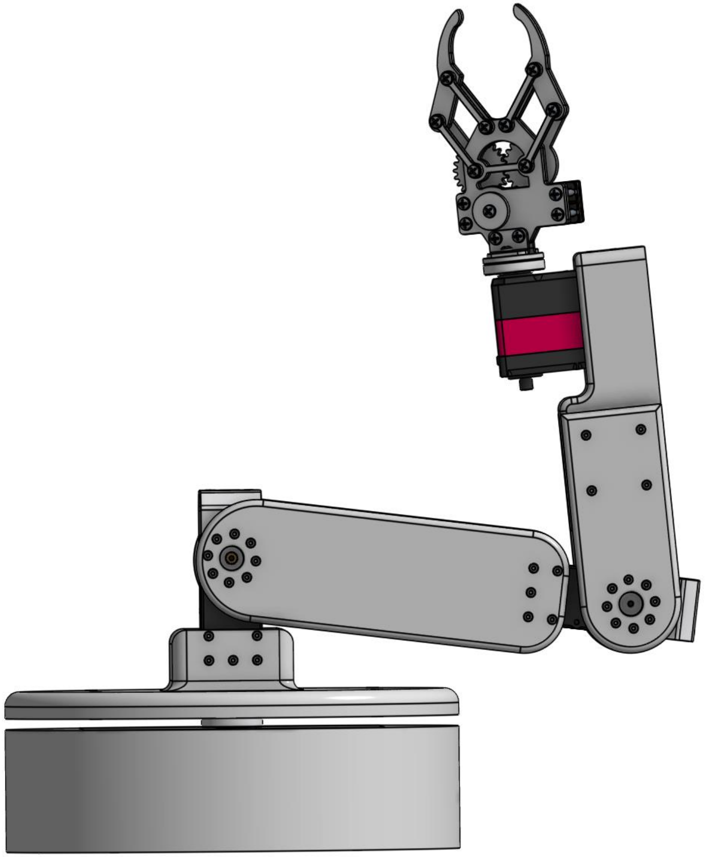
Agenda

- Introductions
- Project Overview
- Mechanical Configuration
- Electrical Configuration
- Coding
- Prototyping
- Demo
- Next Steps
- Q&A

Introduction/Bio

Sergio Garcia	Hector Chavez	Ramiro Garcia
BS Manufacturing Engineering Spring 2026 **Experience** **City of McAllen** ▪ M&SE Administrator 2022-2024 **Skills** C++, Python, 3D Modeling, Circuits, PCB Design.	**BS Manufacturing Engineering** Spring 2026 **Experience** ▪ **Procter & Gamble (2023, 2024, 2025)** **Skills** ABB, FANUC, PLC, Lean Manufacturing, Systems.	**BS Manufacturing Engineering** Spring 2026 **Experience** AI-Driven fused filament fabrication 3d printing process **Skills** Python, SolidWorks, LINDO.

Project Overview



Objectives

- **Design and development of a 5-DOF robotic arm** capable of performing pick-and-place operations **with interchangeable end effectors for glovebox operations.**

Project Overview

Impact

Aid operator in daily tasks for manufacturing setting:

- **Reduce touches (safety)**
- **Automate repetitive tasks (workload)**
- **Increase capability**

Project Overview

Scope

- Design (Custom 3D Model)
- Manufacture (3D Printing)
- System architecture (Components)
- System programming (UI)
- Validation and testing

Initial Product Specifications

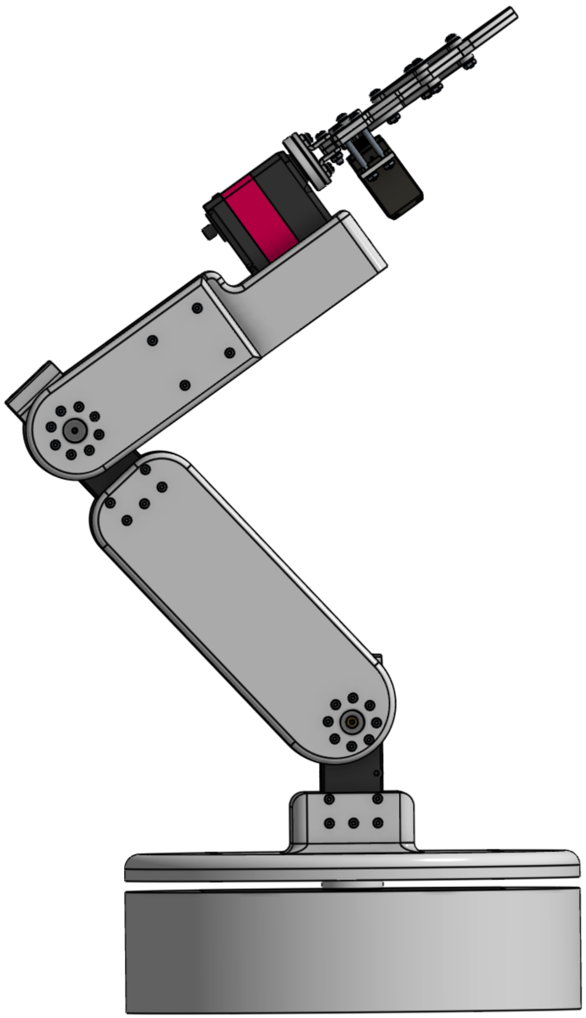
- 1-kg Payload
- Glovebox Dimensions:
  - ~ 1800mm length, 790mm depth, 900 mm height
- Pick-and-place operations: 3 actions and manual operation.

Project Overview

Deliverables

Working robotic arm prototype:

- 3D CAD models
- User Interface (UI)
- Operators' Manual



* As of 11/21

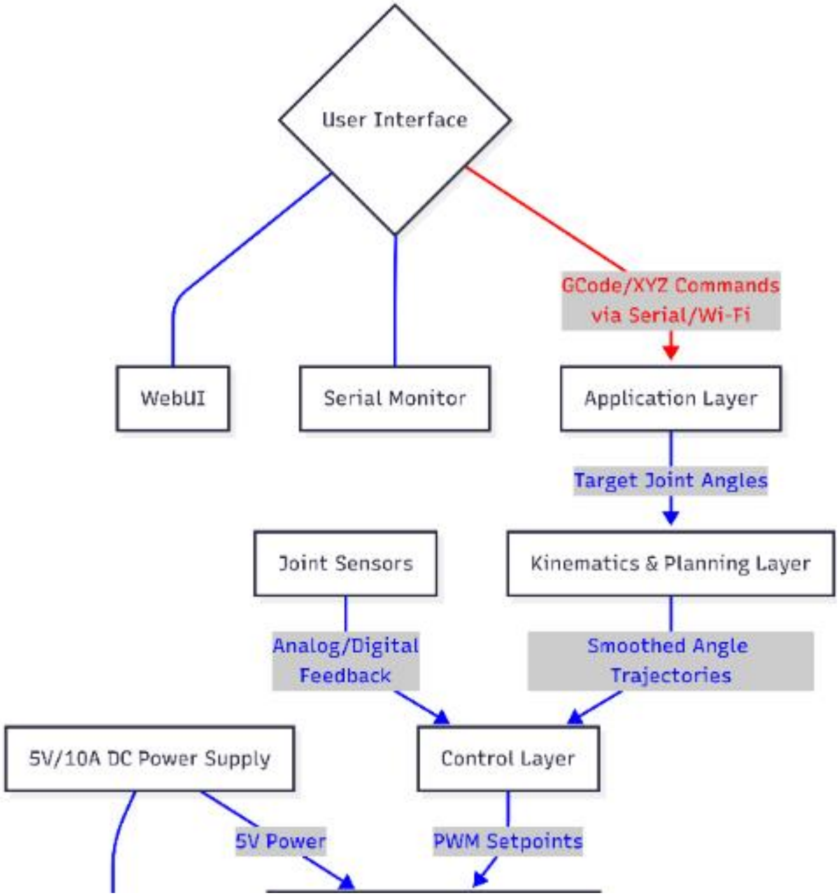
Articulated Robot

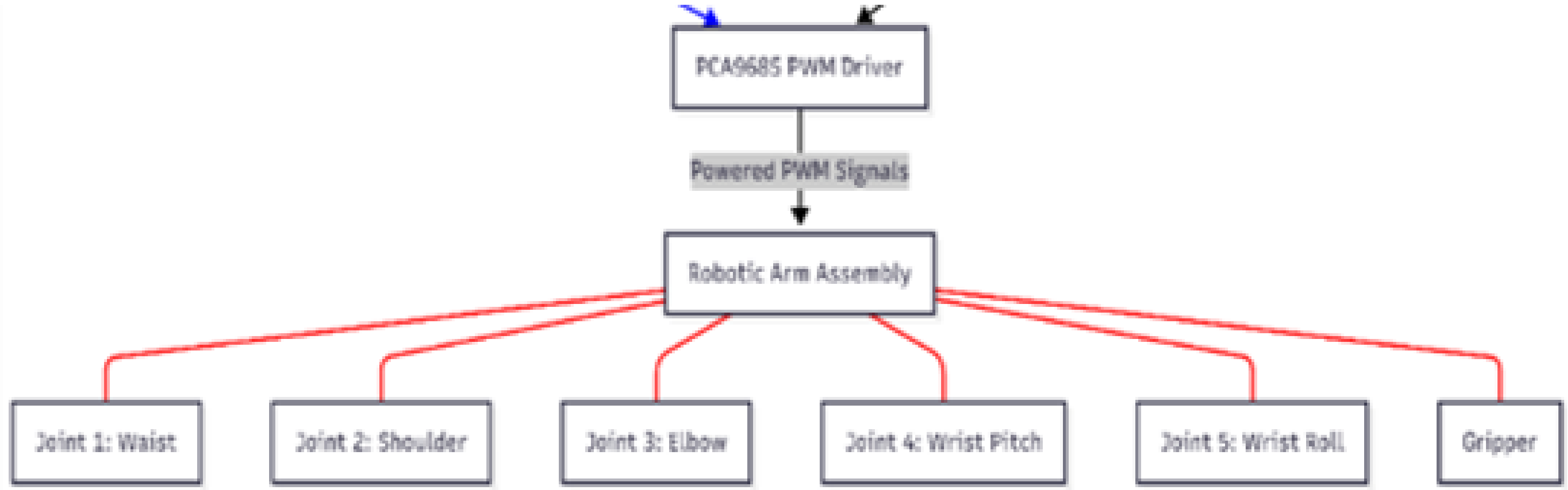
Task	1	2	3	4	5	6	7	8	9	10	12	13	14	15	16
	9/2 -- 9/5	9/8 -- 9/12	9/15 -- 9/19	9/22 -- 9/26	9/29 --10/3	10/6 -- 10/10	10/13 -- 10/17	10/20 -- 20/24	10/27 -- 10/31	11/3 -- 11/7	11/10 -- 11/14	11/17 -- 11/21	11/24 -- 11/29	12/1 -- 12/5	12/8 -- 12/12
Project Initiation/Planning															
Define Scope, Deliverables, Impact															
Define constraints and skills															
Outline ideas for system architecture															
Meet with advisors for feedback/initial guidance on system architecture, first steps					?										
Follow up on feedback from advisors															
Finalize project plan and architecture															
Design & Modeling															
Design waist -- base and waist (J1)															
Design shoulder and elbow (J2, J3)															
Design wrist and gripper (J4, J5)															
Use Gazebo with ROS2 to simulate robotic arm movements and validate print orientations															
Assembly and check part interference															
Finalize (by meeting or email) CAD models and engineering drawings															
Kinematics															
Learn supplementary methematical modeling for kinematics (forward and inverse kinematics)															
Learn about motion planning															
Develop kinematics model															
Systems implementation and coding															
electronics and wiring bench testing															
microcontroller setup for basic servo sweep and control															
code forward kinematics															
code inverse kinematics															
Fabrication															
Create a final BOM															
order components															
3D print parts															
Assemble structure															
Final integration and validation															
integrate mechanics, electronics, code															
develop a method to test systems for robot accuracy and repeatability															
begin pick an place demos															
documentation and final report writing															

COMPLETED	IN PROGRESS	PROJECT COMPLETE

MECHANICAL CONFIGURATION

System Architecture




PCA9685 PWM Driver
Powered PWM Signals
Robotic Arm Assembly
Joint 1: Waist
Joint 2: Shoulder
Joint 3: Elbow
Joint 4: Wrist Pitch
Joint 5: Wrist Roll
Gripper

Mechanical Design

The max reach for our robot should be 0.5 m

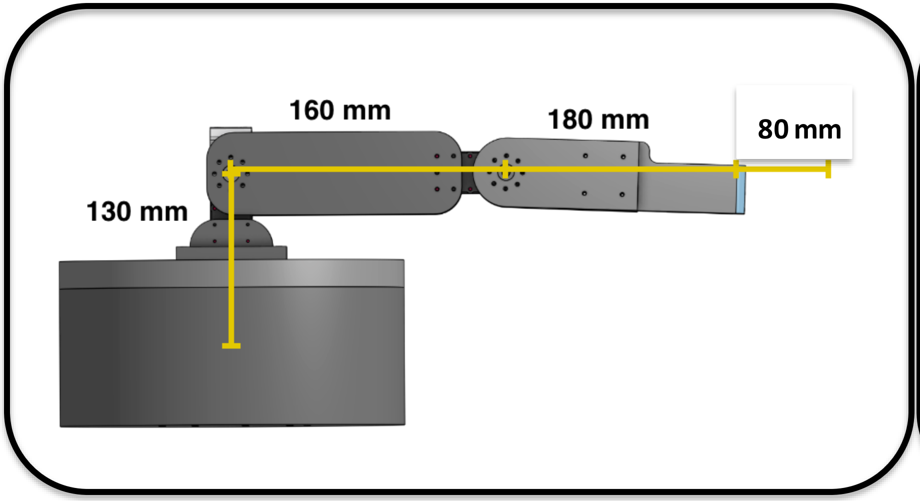
Link / Part	Link Length (m)
Base -> L1 Pivot	0.130
Link 1 (L1)	0.160
Link 2 (L2 + Wrist)	0.180
End Effector	0.8

Link Ratios

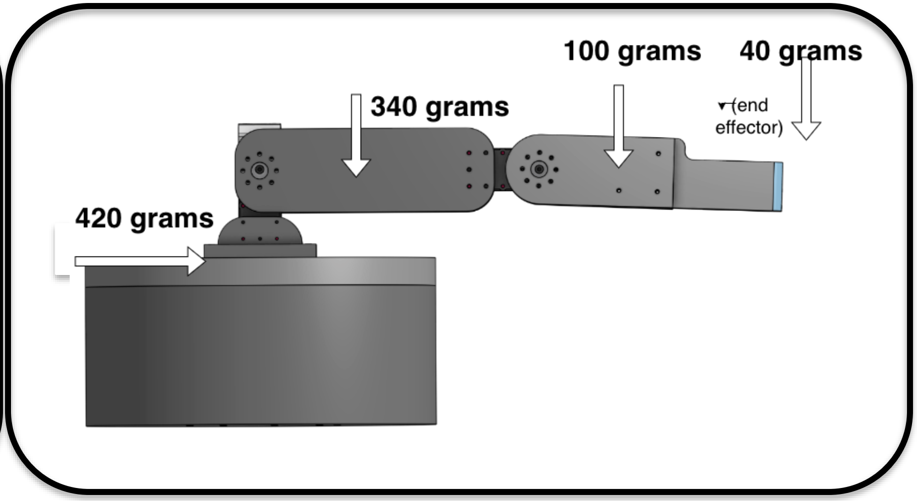
Link Ratio	Ratio
L2 / L1	1.125
End Effector / L2	0.44
End Effector / (L1 + L2)	0.235

- **Allowed for the design our robot's link lengths to our needs while still being close to a small commercial robot**
- **Reduce design risk by validating our geometry against existing robot arms**

Link lengths and Link masses



Link lengths



Link masses

Servo Torque Estimation

The Torque required at each joint is calculated by the function:

$$T_{Req.J} = (g)(m_p + m_{end\ effector} + m_{distal\ link\ masses})(d_j)$$

g = gravitational constant

m_p = mass of payload

$m_{end\ effector}$ = mass of payload

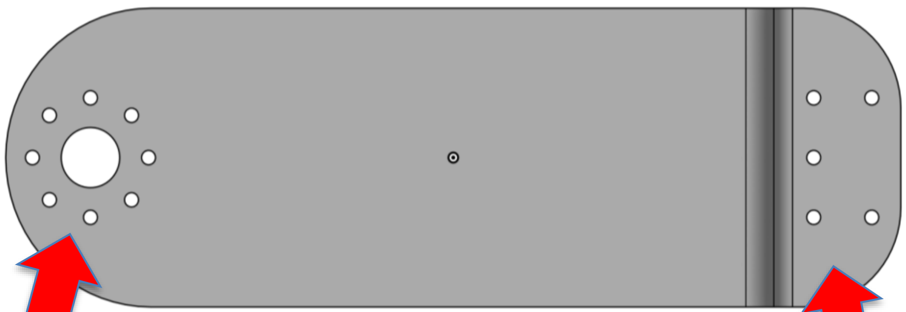
$m_{distal\ link}$ = The sum of link masses, or everything the joint is holding up (all links after it + payload)

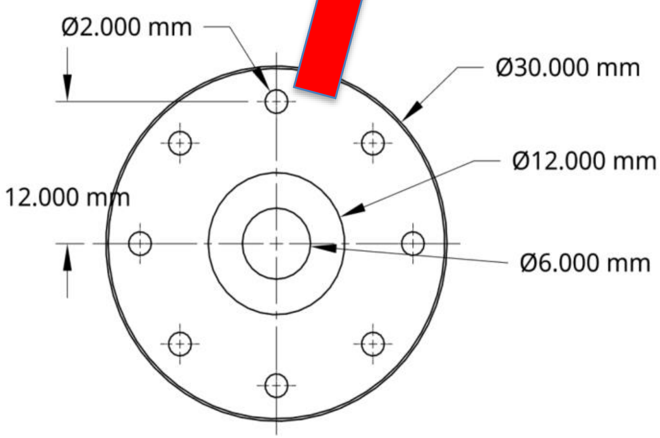
Servo Torque Calculations

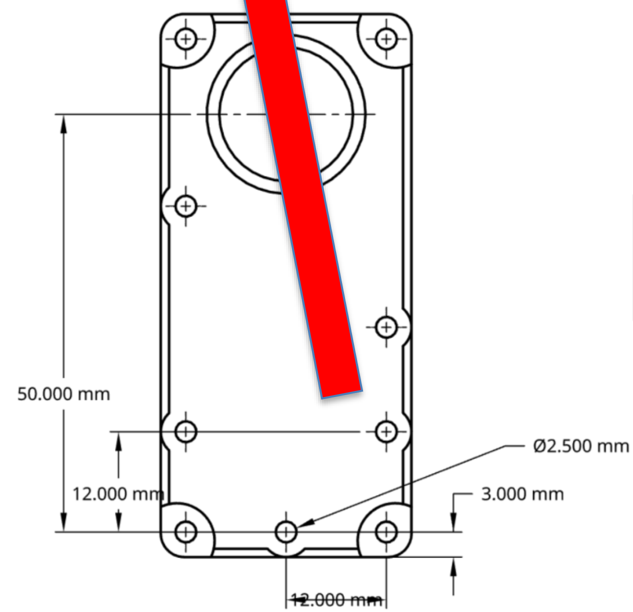
Inputs -- Payload	
Payload mass (kg)	0.5
Gripper (or end effector) mass (kg)	0.05
Inputs -- Link Lengths	
Link 1 length (meters) (base to L1)	0.13
Link 2 length (meters) (Link 1)	0.16
Link 3 length (meters) (Link 2)	0.18
Link 4 length (meters) (end effector)	0.04
Total Link Length	**0.51**
Inputs -- Link Masses	
Link 1 mass + servo 1 (kg)	0.42
Link 2 mass + servo 2 (kg)	0.34
Link 3 mass + servo 3 (kg)	0.1
Link 4 mass + servo 4 (kg)	0.05
Total Link Mass	**0.91**
Inputs -- Other Constants	
Gravity_g (m/s^2)	9.80665
Safety_factor	2
Gear_ratio (motor:output)	1
Drive_efficiency (0-1)	0.9
Conversion_1Nm_to_kgcm	10.19716213

Motor Torque required at joint *n* (kg*cm)	
Motor Torque required at Joint 1 (kg*cm)	**165.4666667**
Motor Torque required at Joint 2 (kg*cm)	**87.82222222**
Motor Torque required at Joint 3 (kg*cm)	**34.22222222**
Motor Torque required at Joint 4 (kg*cm)	**5.333333333**

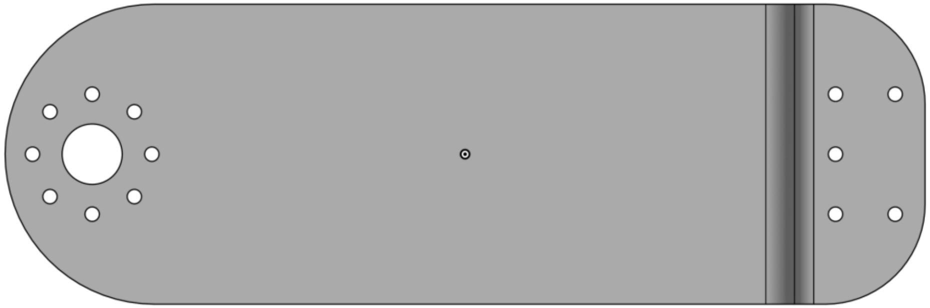
3D Modeling – Assembly

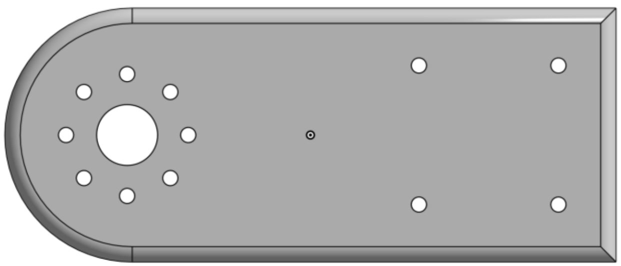


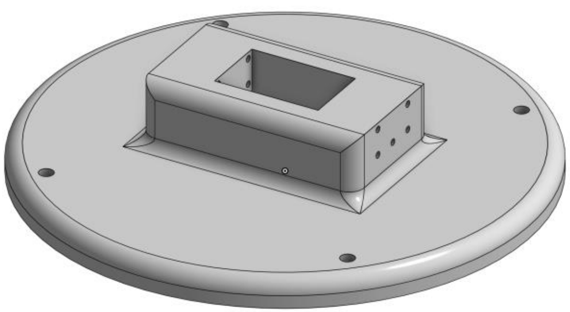


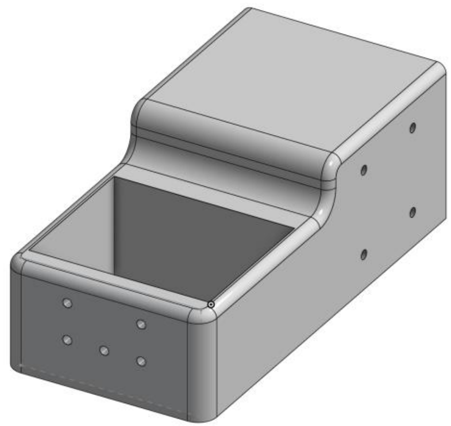


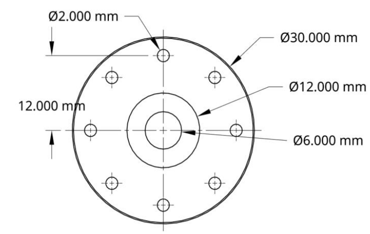
3D Modeling – Assembly

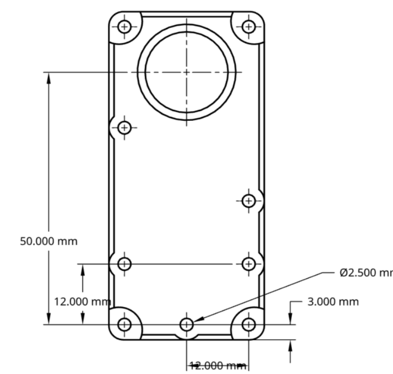




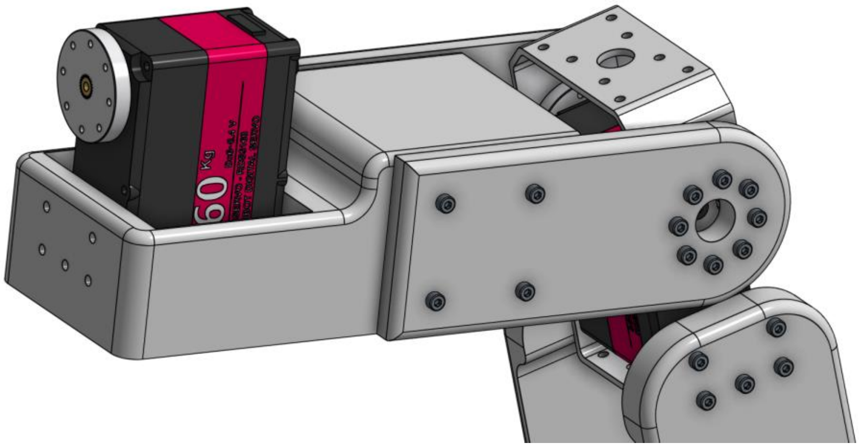


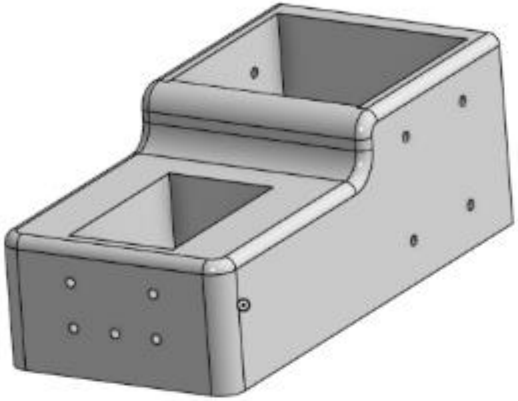






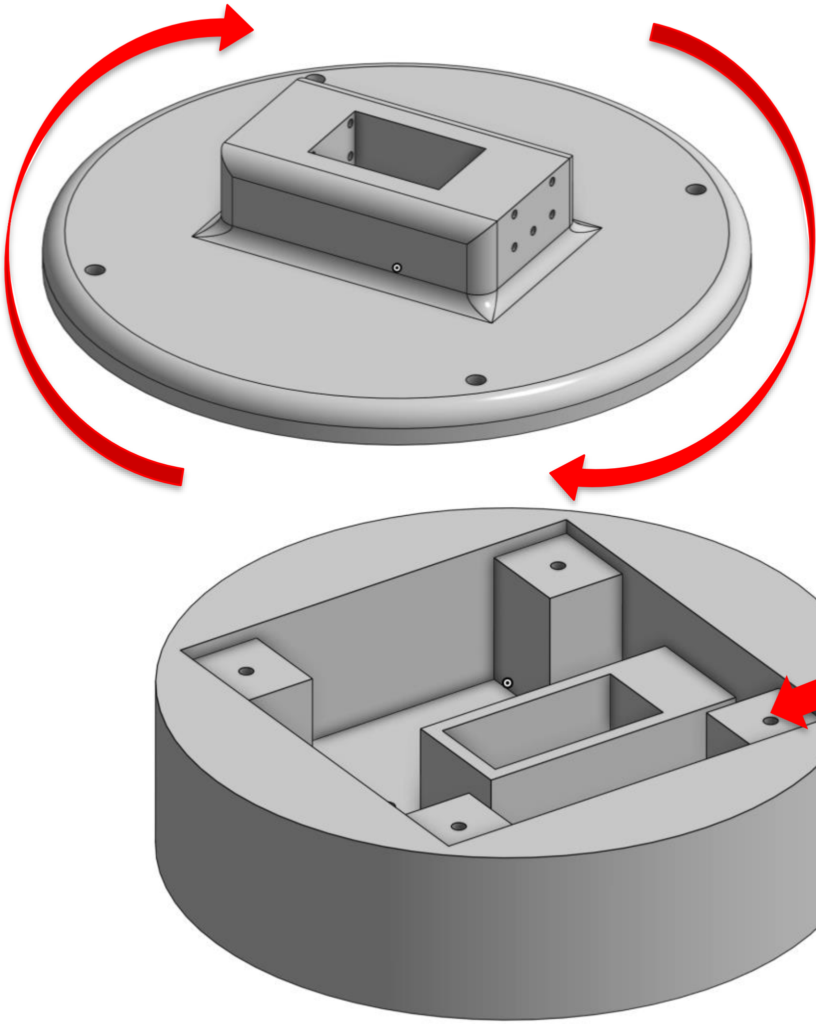
Link 2 and Wrist

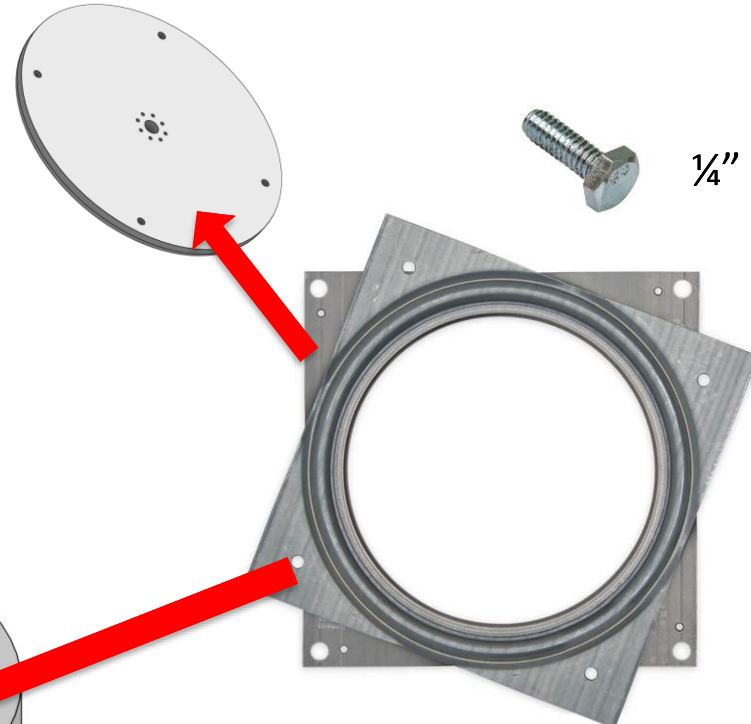




Wrist supports link 2 while carrying payload by **increasing the area moment of inertia, reducing bending.**

Base + Base plate





¼” screws

3/16” screws

Printing

Selected material – **PLA-CF**

Mechanical Properties:

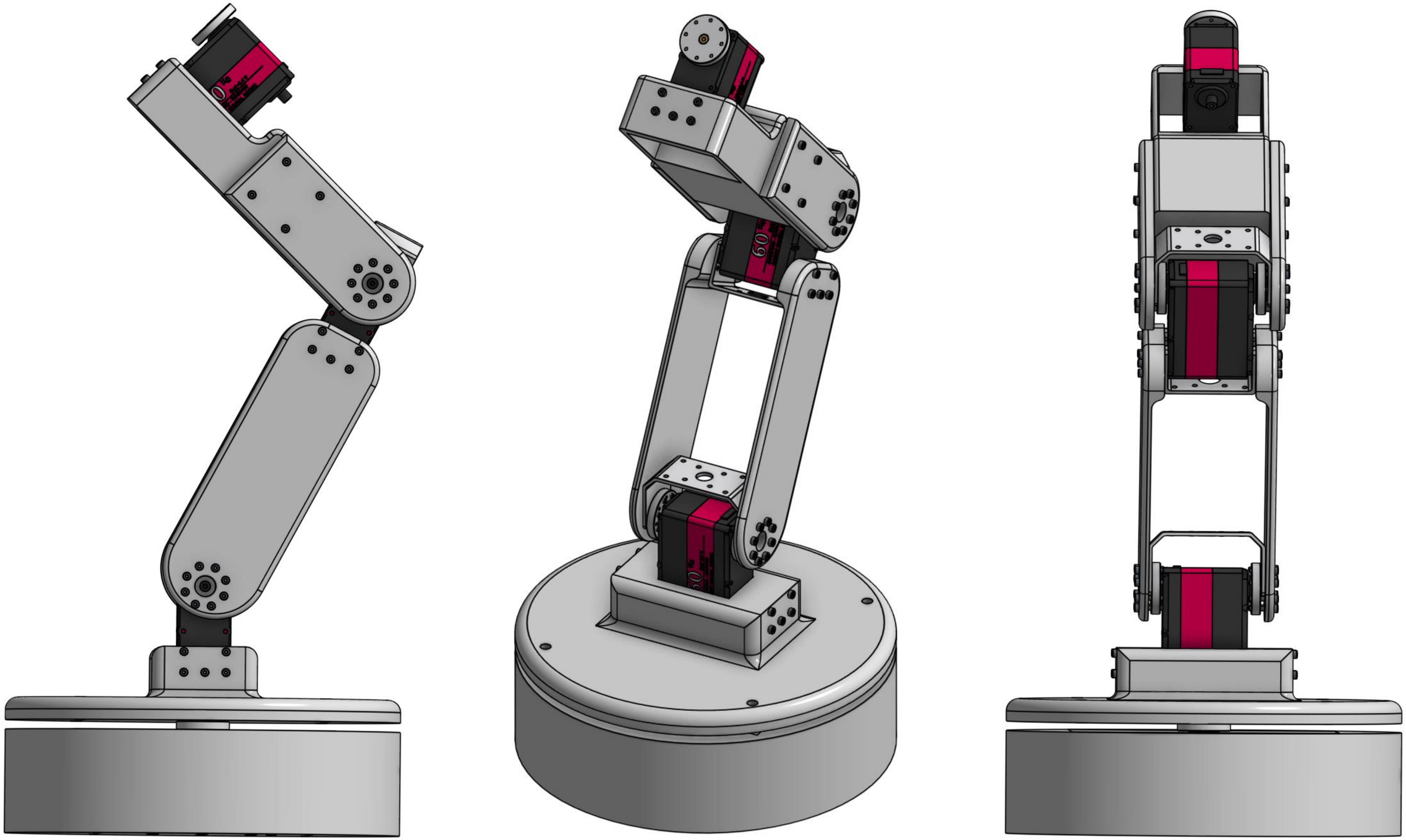
- Density – 1.22 g/cm^3 (lighter than aluminum)
- High flexural (bending) modulus (3950 MPa)
- High tensile strength (38 MPa)

Machines Used

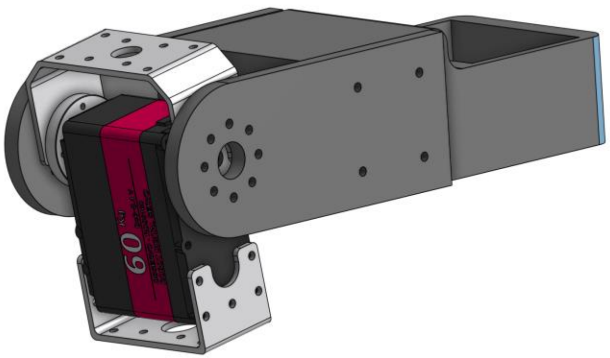
Bambu Lab X1C

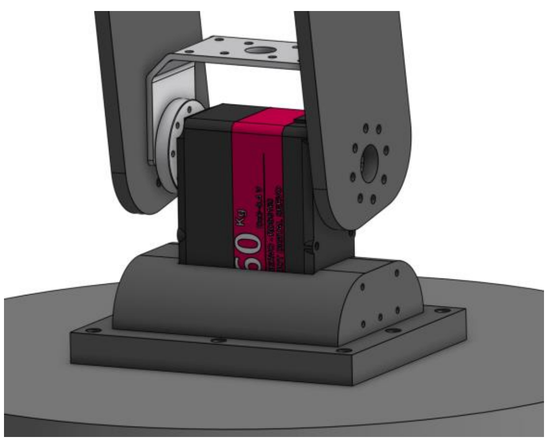
- **Build volume:** 256 × 256 × 256 mm.
- **Features:** Multi-material / multi-color printing (up to 16 colors/materials with AMS), dual auto bed leveling (Lidar + force sensor), filament run-out detection, “spaghetti” failure detection & auto-pause, power-loss recovery

3D Modeling – Assembly

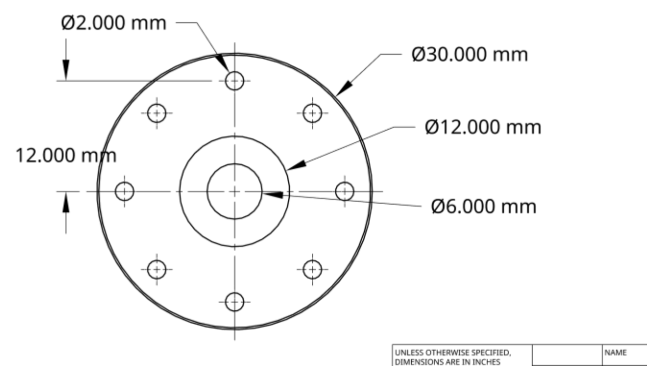


Holding the assembly together





- The screw holes for the links are designed around the servo motors rotary discs
- M2.5 screws hold the links to the brackets of the servo motor
- The same screw also holds the rotary disc, bracket, and outermost link together

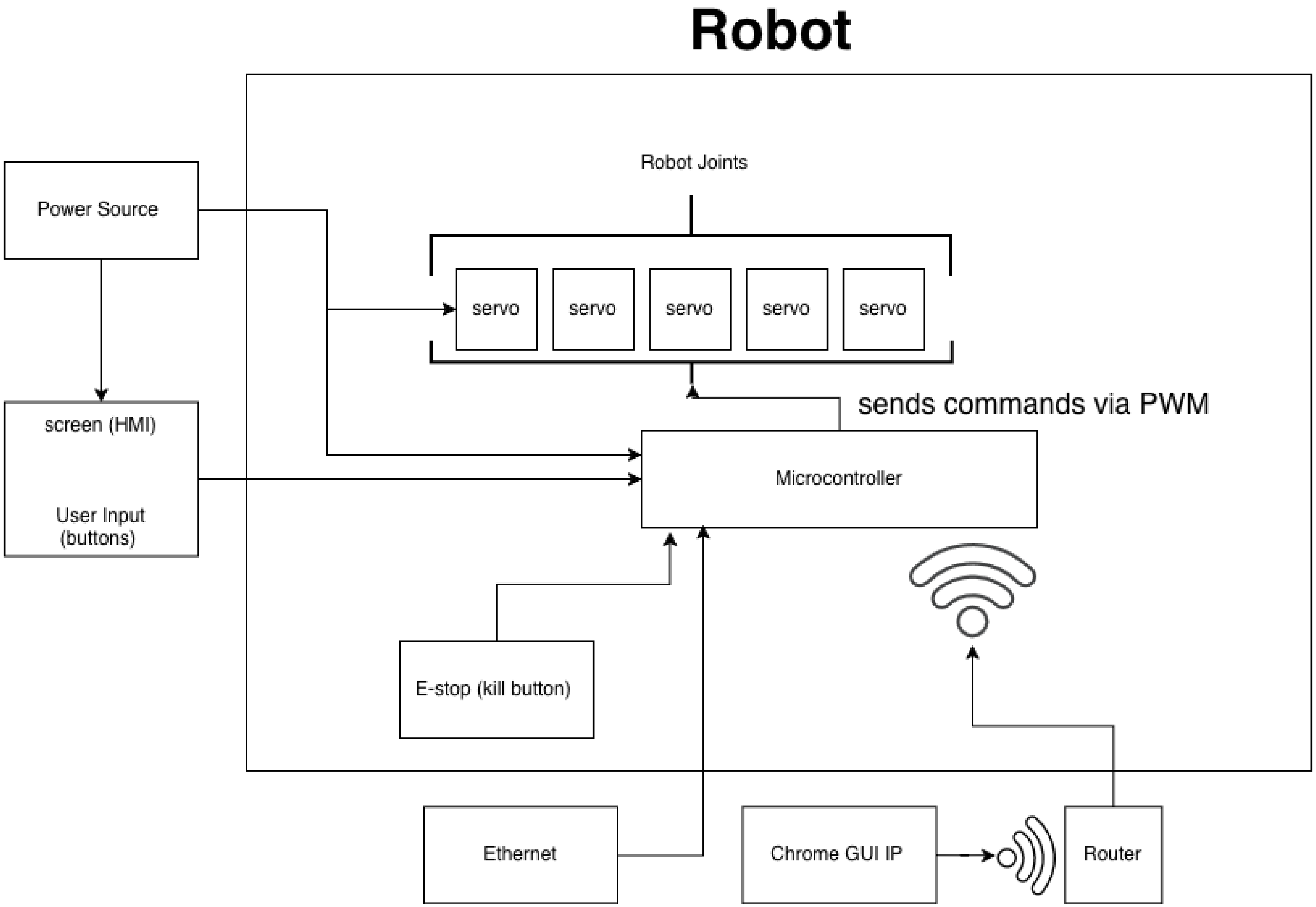


Mechanical – Lessons Learned

Objectives	Deliverables	Learnings
System Architecture	Complete system layout with component roles	Control layers, use of FWK and IK in robotics
Components Specifications	Identified specific components and created a Bill of Materials (BOM)	Servo sizing, Microcontroller resolution and impact, PWM signal
3D Modeling	Complete CAD model of the robot	Tolerancing, design-for-assembly & rapid prototyping

ELECTRICAL CONFIGURATION

Components Diagram



Servo Selection

- Operating Voltage:
  - 24V
- Idle Current:
  - 5mA at 24V
- Operating Speed:
  - 0.21 sec/60° at 24V
- Stall Torque:
  - 170kg·cm at 24V

Power Supply Requirements for Servo operation

- **Total Stall current:**
  - 4.2A * 5 Servos = **21 A**

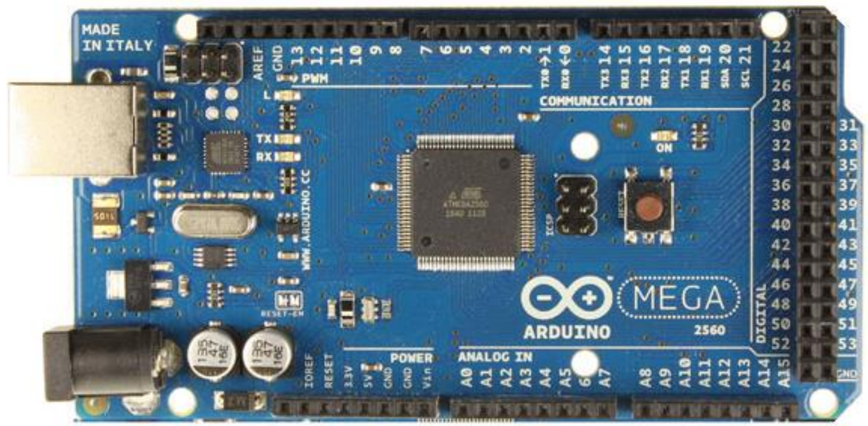
- **Power at 24V during a total stall:**
  - 24V * 21 A = **504 W**

Power Supply

Selected Power Supply:
24V 50A 1200W

Custom power supply housing

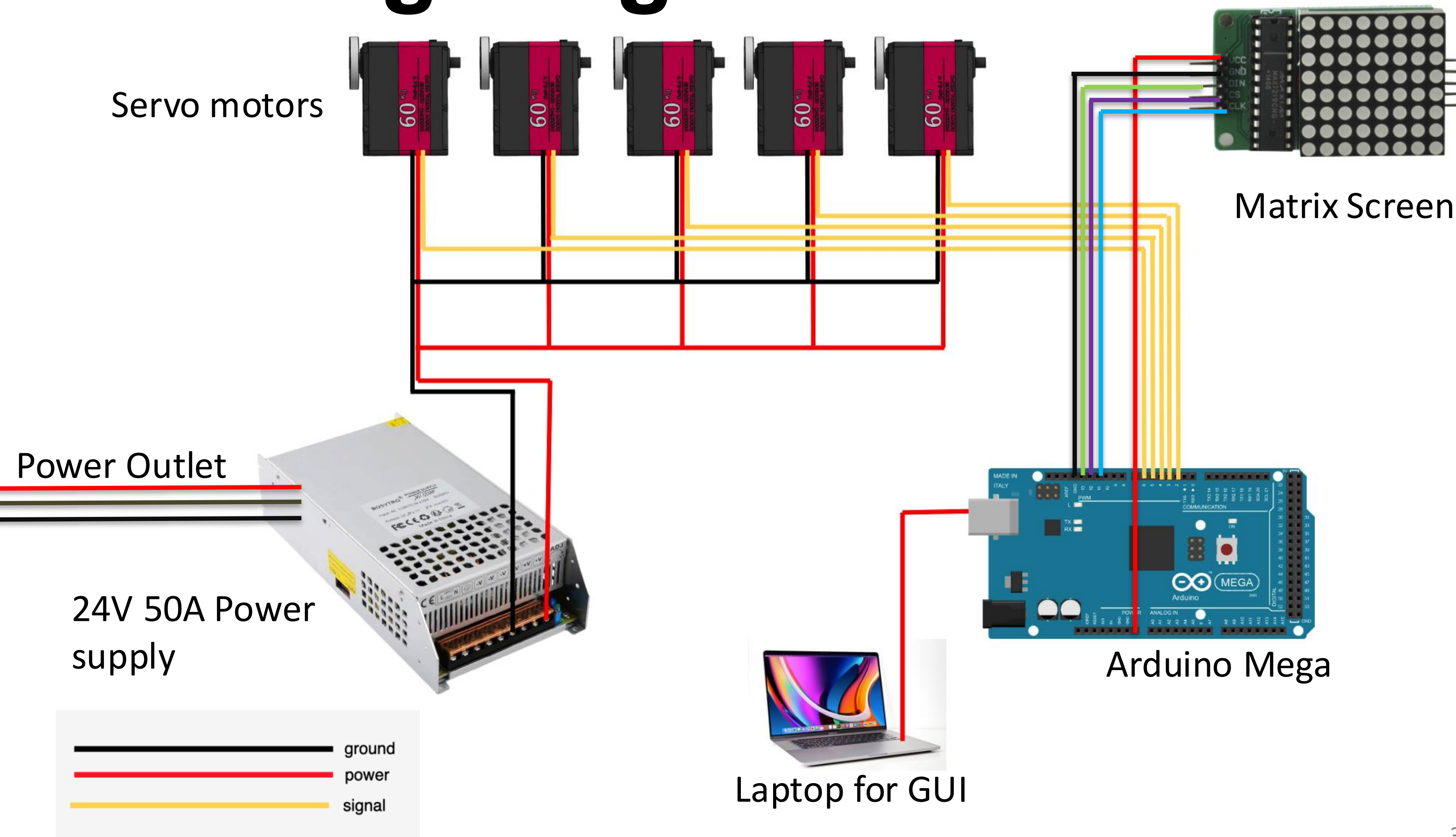
Microcontroller



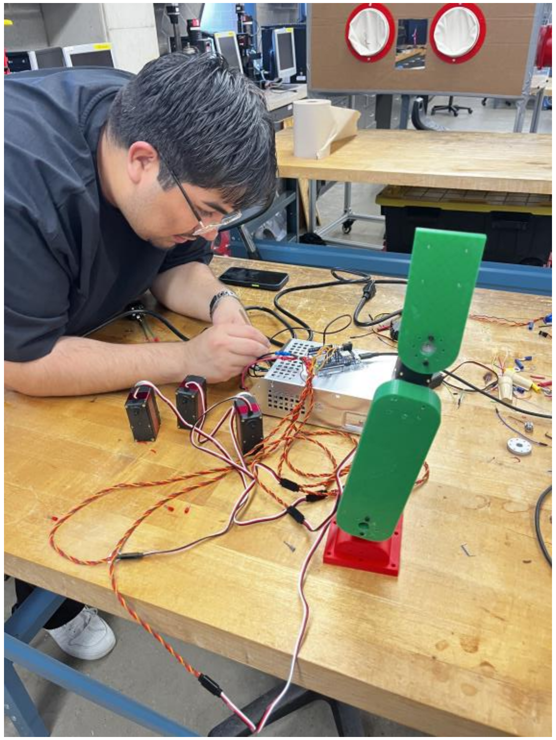
Arduino Mega

- Arduino Mega uses **8-bit PWM** (0–255 steps)
  - 270º / 255 steps = 1.05º per step

Wiring Diagram



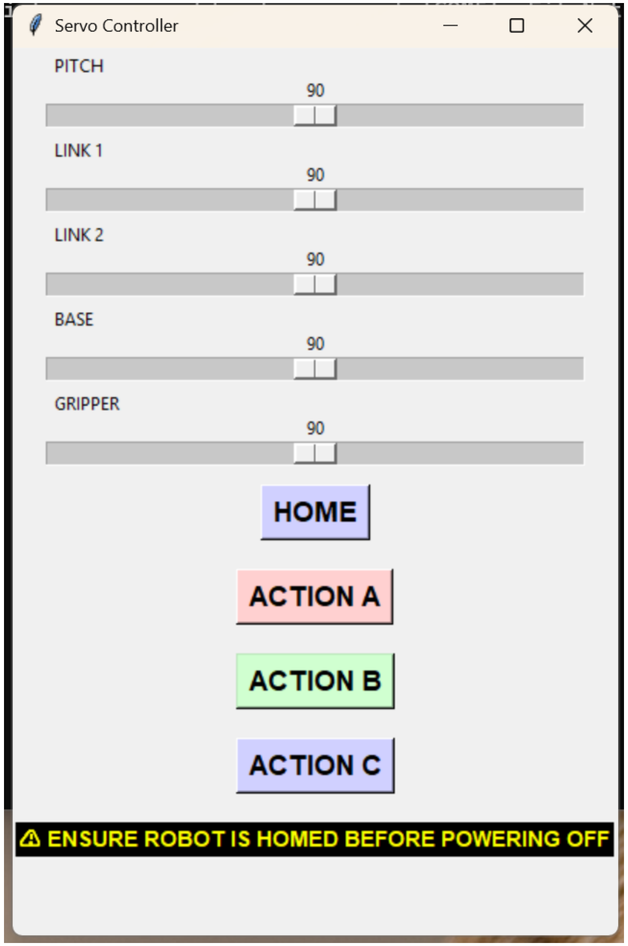
Electrical – Lessons Learned

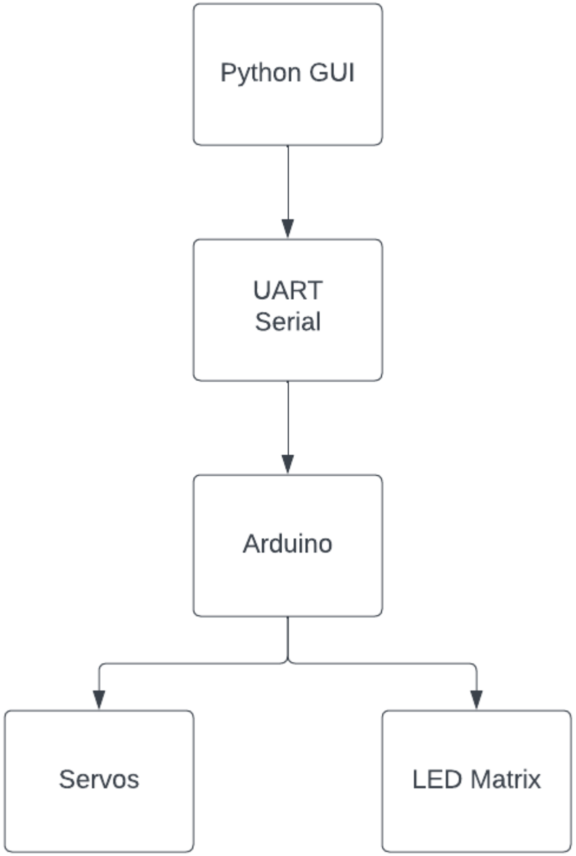


Objectives	Deliverables	Learnings
Servo Powering	Stable multi-servo wiring	Current draw, power limits, shared ground importance
Signal Routing	Organized wiring layout	Cable management, avoiding interference
Power Supply Setup	External power integration	System voltage needs and current draw

PROGRAMMING

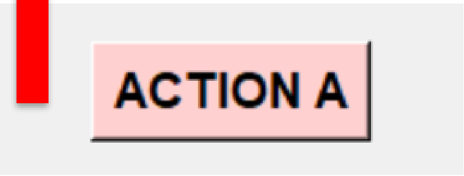
GUI AND SYSTEM DIAGRAM

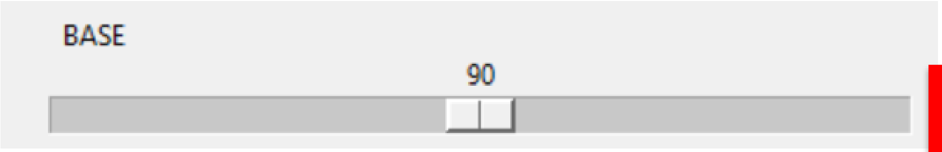




TKINTER INTERFACE

```python
btn_actionA = tk.Button(root, text="ACTION A", font=("Arial", 14, "bold"), command=actionA, bg="#ffd0d0")
btn_actionA.pack(pady=10)

btn_actionB = tk.Button(root, text="ACTION B", font=("Arial", 14, "bold"), command=actionB, bg="#d0ffd0")
btn_actionB.pack(pady=10)

btn_actionC = tk.Button(root, text="ACTION C", font=("Arial", 14, "bold"), command=actionC, bg="#d0d0ff")
btn_actionC.pack(pady=10)
```



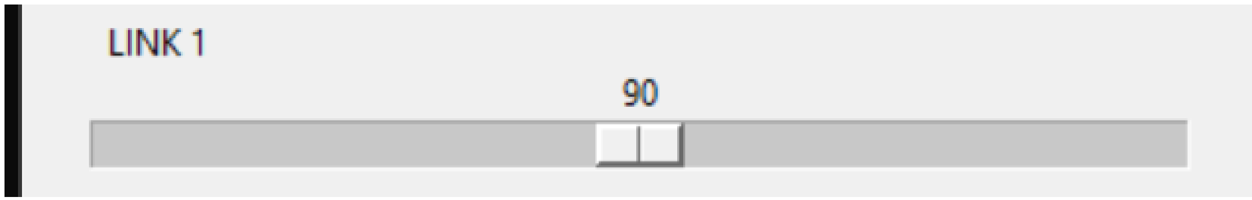


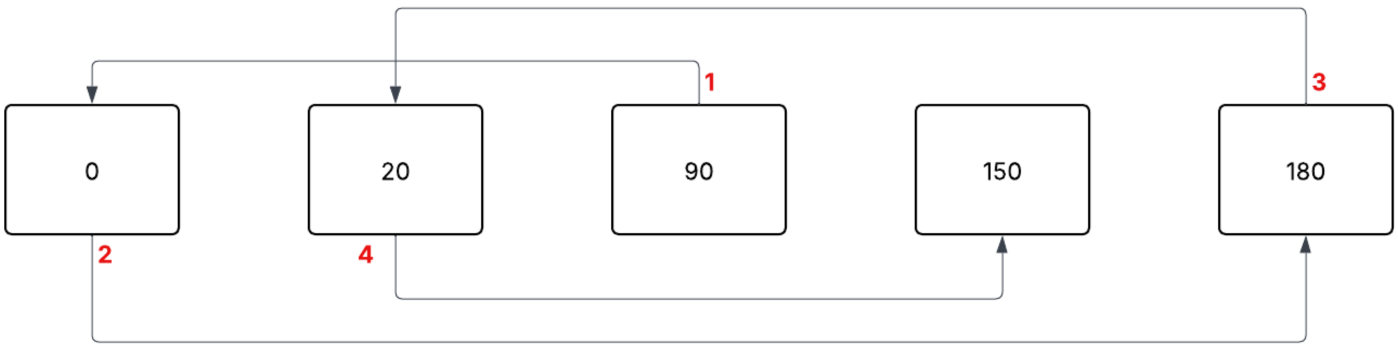

```python
s1 = tk.Scale(root, from_=0, to=180, orient="horizontal", label="PITCH", command=slider_changed); s1.set(DEFAULT_ANGLE)
s2 = tk.Scale(root, from_=0, to=180, orient="horizontal", label="LINK 1", command=slider_changed); s2.set(DEFAULT_ANGLE)
s3 = tk.Scale(root, from_=0, to=180, orient="horizontal", label="LINK 2", command=slider_changed); s3.set(DEFAULT_ANGLE)
s4 = tk.Scale(root, from_=0, to=180, orient="horizontal", label="BASE", command=slider_changed); s4.set(DEFAULT_ANGLE)
s5 = tk.Scale(root, from_=0, to=180, orient="horizontal", label="GRIPPER", command=slider_changed); s5.set(DEFAULT_ANGLE)
```

SERVO MOTION SEQUENCE

```python
def actionB():
    global ready
    ready = True

    display("B") # GO CENTER
    move_all_slow([90, 156, 90, 100, 90])
    time.sleep(1)

    display("B") # GO HOME
    move_all_slow([90, 90, 90, 90, 90])
    time.sleep(1)

    display("B") # GO CENTER
    move_all_slow([90, 156, 90, 100, 90])
    time.sleep(1)

    display("SMILE")
    move_all_slow([90, 90, 90, 90, 90])
```

SLIDER MOTION LOGIC






```
const int STEP_SIZE = 2;
const unsigned long STEP_DT = 20;
```

R: 90 → 92 → 94 → 96 →

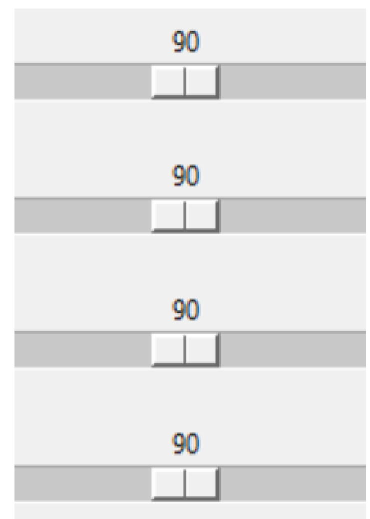
L: 90 → 88 → 86 → 84 →

LED MATRIX

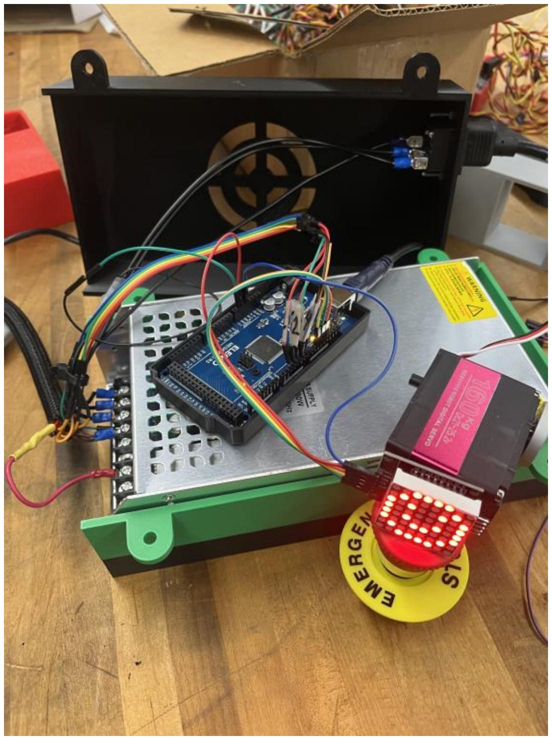
```
// LETTER A
byte letterA[8] = {
  B00011000,
  B00100100,
  B01000010,
  B01000010,
  B01111110,
  B01000010,
  B01000010,
  B01000010
};
```

INITIAL POSITION LOGIC

- Servos power on in their last physical position
- First PWM forces immediate movement
- Centering reduces startup torque
- Provides a consistent starting pose



PROGRAMMING – LESSONS LEARNED

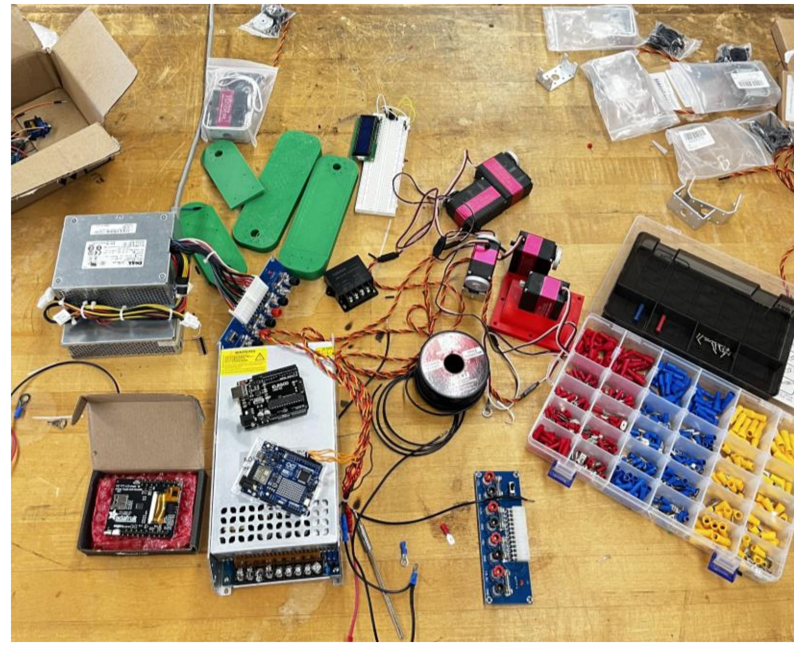


Objectives	Deliverables	Learnings
Build control logic	Servo motion functions (smooth move, presets)	Multi-servo timing, synchronization behavior
Implement smooth servo motion	Smooth-movement parameters (STEP + STEP_DT)	Motion planning
Integrate GUI–Arduino communication	Python Tkinter GUI with sliders + buttons	UART behavior, timing delays, command handling

PROTOTYPING

Assembling the Prototype

- Spent extra time in the initial design phase
- Planned a design that's easy to assemble and print
- Optimized for rapid prototyping
- "Measure twice, cut once" approach

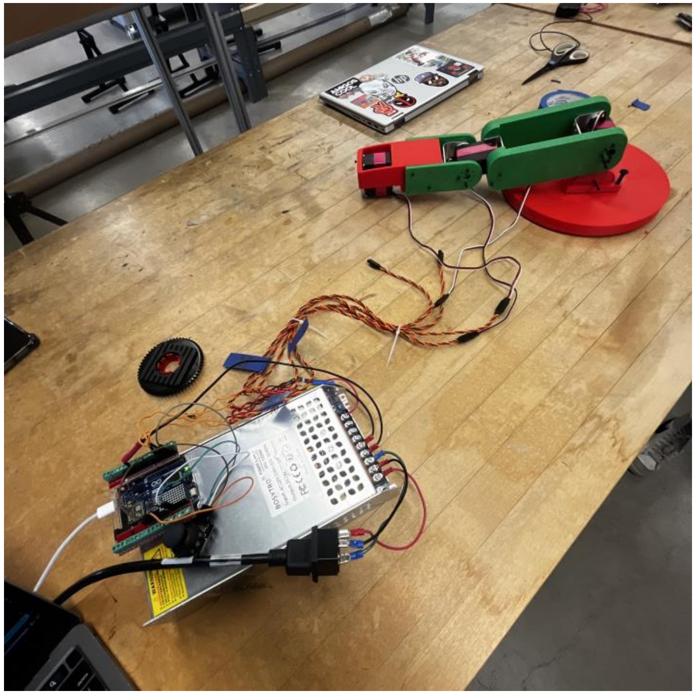


UTRGV Manufacturing & Industrial Engineering

Prototype timeline

In Collaboration With: Los Alamos National Laboratory TechSource Savannah River Nuclear Solutions

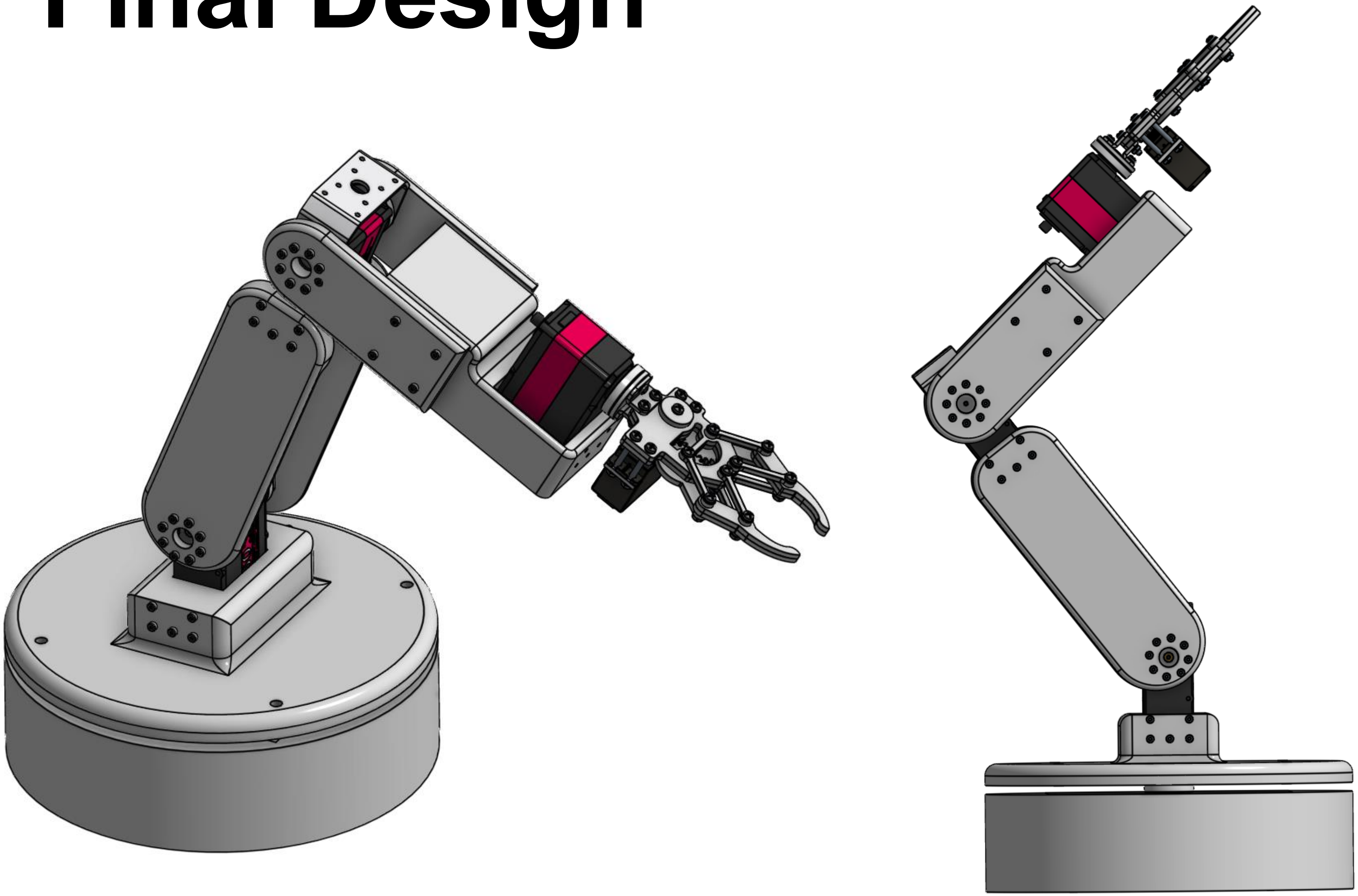
Prototyping



Objectives	Deliverables	Learnings
Assemble first prototype	Working prototype in a pick and place demo, operator and automatic mode configured	▪ Rapid prototyping ▪ Remember to check all wiring
Validation and testing	Repeatability of (X)	

FINAL DESIGN

Final Design



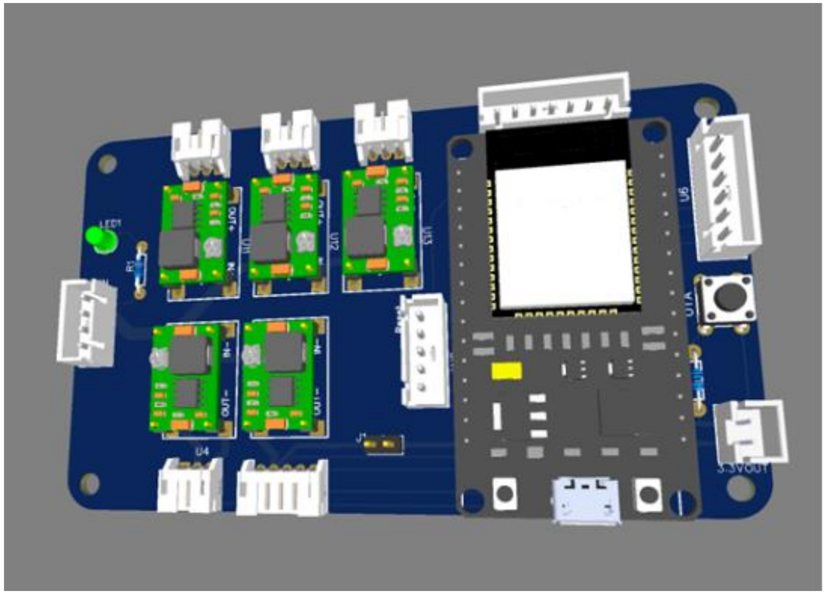
NEXT STEPS

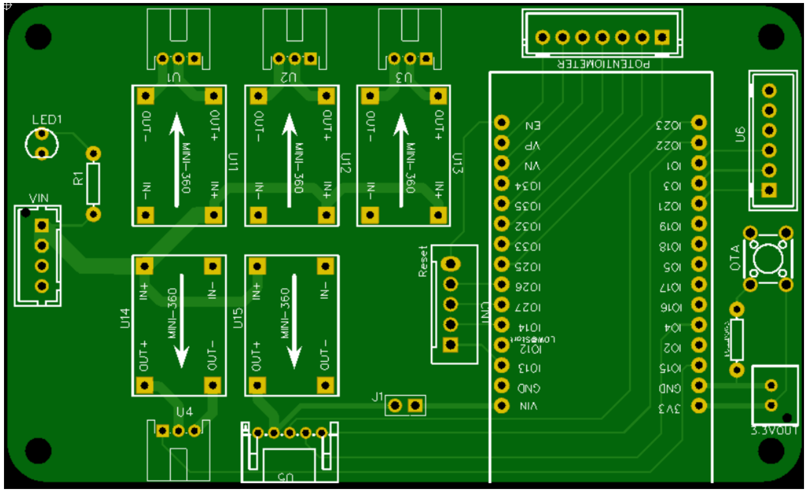
Gazebo Simulation

- Import the robot model into Gazebo for testing
- Simulated robot paths and motions
- Validated servo angles for each position

GAZEBO

Custom PCB


LED1
U11
U12
U13
U6
OTA
U4
J1
3.3VOUT


LED1
R1
VIN
U1
U2
U3
U4
U5
U6
U11
U12
U13
U14
U15
MINI-360
OUT+
OUT−
IN+
IN−
Reset
CN1
J1
OTA
3.3VOUT
EN
VP
VN
IO34
IO35
IO32
IO33
IO25
IO26
IO27
IO14
IO12
IO13
GND
VIN
IO23
IO22
IO1
IO3
IO21
IO19
IO18
IO5
IO17
IO16
IO4
IO2
IO15
3V3

Automatic Tool Changer

THANK YOU FOR YOUR TIME

Questions?